

Sme Note

Acknowledge: Sample Space: Ω Random Variables: $\mathbf{Y} = (Y_1, \dots, Y_n)$ Sample Data: $\mathbf{y} = (y_1, \dots, y_n)$ $f_{\mathbf{Y}}(\mathbf{y}; \theta) = \prod_i f_{Y_i}(y_i; \theta)$, if Y_i are independent Distribution Function: $F_{\mathbf{Y}}(\mathbf{y}) = \mathbb{P}(Y_1 \leq y_1, \dots, Y_n \leq y_n)$
Parameter Space: Θ CDF: $F_Y(y)$ PDF/PMF: $f_Y(y)$ or $p_Y(y)$

1 Basic Knowledge

Def of expectation (Moments): For any well-behaved function h on $\mathbb{R}$. $\mathbb{E}[h(Y)] := \sum_i h(y_i) \mathbb{P}(Y = y_i)$ $\mathbb{E}[h(Y)] := \int_{\mathbb{R}} h(y) f_Y(y) dy$

Def of Model: A collection of distributions indexed by a parameter $\{F_{\mathbf{Y}}(\cdot|\theta); \theta \in \Theta\}$ is model.

Likelihood Function: $L(\theta; \mathbf{y}) = f_{\mathbf{Y}}(\mathbf{y}; \theta)$ **MLE:** $\hat{\theta}(\mathbf{y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{y})] = [U(\hat{\theta}) = 0]$ ps: MLE has invariant property

(Strong) Likelihood Principle: $L(\theta; x) \propto L(\theta; y)$ [Strong: $L_f(\theta; x) \propto L_g(\theta; z)$; f, g are statistical model] $\Rightarrow$ Same Inferential Conclusions.

Def of Identifiability: A model is identifiable if $f(y; \theta_1) = f(y; \theta_2), \forall y$ then $\theta_1 = \theta_2$ (i.e. $\text{map} : \theta \rightarrow f(y; \theta)$ is 1-1)

Linear Algebra Knowledge: If A is symmetric matrix, with eigenvalues λ_n then $A + aI$ have eigenvalues $\lambda_n + a$. $\| \det(M) = \prod \lambda_n$ eigenvalues of $X^T X \geq 0$.

· Method: 1.By def $\Rightarrow$ ratio not depend on $y \Rightarrow \theta_1, \theta_2$ relationship. (with any additional constraint) 2.MLE not unique $\Rightarrow$ not identifiable 3.Give an example θ 4.Fisher information

2 Properties of the Likelihood and the MLE

Positive [Semi-]definite matrix M: $\forall \mathbf{z} \in \mathbb{R}^n \setminus \mathbf{0}, \mathbf{z}^T \mathbf{M} \mathbf{z} > 0 [\geq 0]$ $\Leftrightarrow (> 0)^1$ n positive eigenvalues. 2 full rank. 3 invertible

Score Func / Vector U	Single: $U(\theta, \mathbf{Y}) = \ell'(\theta; \mathbf{Y})$	$\mathbb{E}[U] = 0$	$Var[U] = I(\theta)$
	Multiple: $\mathbf{U}(\theta) = [\partial_{\theta_1} \ell(\theta; \mathbf{Y}), \dots, \partial_{\theta_p} \ell(\theta; \mathbf{Y})]^T \in \mathbb{R}^p$	$\mathbb{E}[\mathbf{U}] = \mathbf{0}$	$Var[\mathbf{U}] = I(\theta)$ $I_{j,k} = -Cov(\frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_j}, \frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_k}) = -\mathbb{E}[\frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_j} \frac{\partial \ell(\theta, \mathbf{Y})}{\partial \theta_k}]$
CLT of U	Single: $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, I(\theta_0)), n \rightarrow \infty$		
	Multiple: $\mathbf{U}(\theta_0; \mathbf{Y}) \sim \mathcal{N}_p(\mathbf{0}, I(\theta_0)), n \rightarrow \infty$	Also: $\mathbf{U}(\theta_0; \mathbf{Y})^T I^{-1}(\theta_0) \mathbf{U}(\theta_0; \mathbf{Y}) \sim \chi_p^2$	
Fisher Information I	Single: Observed: $J(\theta, \mathbf{y}) = -U'(\theta, \mathbf{y}) = -\ell''(\theta; \mathbf{y})$	Expected: $I(\theta) = -\mathbb{E}[\ell''(\theta; \mathbf{Y})]$ $\ $ Unit: $i(\theta) = I(\theta)/n = -\mathbb{E}[\ell''(\theta; Y_i)]$	
	Multiple: Observed: $\mathbf{J}(\theta; \mathbf{y}) \in \mathbb{R}^{p \times p}, J_{j,k} = -\frac{\partial^2 \ell(\theta, \mathbf{y})}{\partial \theta_j \partial \theta_k}$	Expected: $I(\theta) \in \mathbb{R}^{p \times p}, I_{j,k} = -\mathbb{E}[\frac{\partial^2 \ell(\theta, \mathbf{Y})}{\partial \theta_j \partial \theta_k}]$ ps: $I(\theta)$ always positive semi-definite	
$I \uparrow \Rightarrow \theta$ 对 θ 越敏感; 信息量大; 参数估计更精确		A model is <i>identifiable</i> iff $I(\theta)$ is positive definite. ps: (一维) identifiable if $I(\theta) \neq 0$	
CLT of MLE	Single: $I(\theta_0) < \infty \Rightarrow \hat{\theta} \sim \mathcal{N}(\theta_0, I^{-1}(\theta_0)), n \rightarrow \infty$	$\widehat{Var}(\hat{\theta}) \approx I^{-1}(\hat{\theta})$	$ESE(\hat{\theta}) \approx \sqrt{I^{-1}(\hat{\theta})}$ estimated standard error
CLT of I	Multiple: $I(\theta_0) < \infty \Rightarrow \hat{\theta} \sim N_p(\theta_0, I^{-1}(\theta_0)), n \rightarrow \infty$	Also: $(\hat{\theta} - \theta_0)^T I(\theta_0) (\hat{\theta} - \theta_0) \sim \chi_p^2$	

2.1 For Single Parameter

*Random Variable / Function: $L(\theta; \mathbf{Y})$ is random function (NOT pdf). For fixed θ , it's r.v. $\hat{\theta}(\mathbf{Y}) = \text{ArgMax}_{\theta \in \Theta} [L(\theta; \mathbf{Y})]$ is r.v.

ps: I^{-1}, J^{-1} 是指其负一次方, 不是反函数

True Parameter: θ_0

Name	Test Statistic	Hypothesis	CI	Other
Score Test	Given $U(\theta_0; \mathbf{Y}) \sim \mathcal{N}(0, I(\theta_0))$ $T_{Score}(\mathbf{Y}) = \frac{U(\theta_0)}{\sqrt{I(\theta_0)}} \sim \mathcal{N}(0, 1)$	$H_0 : \theta = \theta_0$ 不用 MLE, 泰勒展开	$\left\{ \theta : \frac{U^2(\theta)}{I(\theta)} < z_{\alpha, 1}^2 \right\}$	or $T_{Score}^2(\mathbf{Y}) \sim \chi_1^2$ $T_{Score}(\mathbf{Y}) \approx \frac{U(\theta_0)}{\sqrt{I(\theta_0)}}, \frac{U(\theta_0)}{\sqrt{J(\theta_0)}}, \frac{U(\theta_0)}{\sqrt{J(\hat{\theta})}}$
Wald Test or Maximum Likelihood (ML) Test	$T_{Wald}(\mathbf{Y}) = \sqrt{I(\theta_0)}(\hat{\theta} - \theta_0) = \frac{\hat{\theta} - \theta_0}{\sqrt{I^{-1}(\theta_0)}}$ $T_{Wald}(\mathbf{Y}) \sim \mathcal{N}(0, 1)$	$H_0 : \theta = \theta_0$ 构建 CI	$\left\{ \theta : (\hat{\theta} - \theta)^2 I(\theta) < z_{\alpha, 1}^2 \right\}$	or $T_{Wald}^2 \sim \chi_1^2$; $n \rightarrow \infty$ $T_{Wald} \approx T_{Score}$ $T_{Wald}(\mathbf{Y}) \approx \frac{(\hat{\theta} - \theta_0)}{\sqrt{I^{-1}(\hat{\theta})}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{J^{-1}(\theta_0)}}, \frac{(\hat{\theta} - \theta_0)}{\sqrt{J^{-1}(\hat{\theta})}}$
log-Likelihood Ratio Test	$T_{LR}(\mathbf{Y}) = 2(\ell(\hat{\theta}) - \ell(\theta_0)) \sim \chi_1^2$ as $n \rightarrow \infty$ $LR = L(\theta_0)/L(\hat{\theta}), T_{LR} = -2\ln(LR)$	$H_0 : \theta = \theta_0$ 最小 CI, Most Powerful	$\left\{ \theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha, 1}^2 \right\}$	ps: $2(\ell(\hat{\theta}) - \ell(\theta_0)) \approx (\theta_0 - \hat{\theta})^2 J(\hat{\theta})$
$T_{Score} \approx T_{Wald} \approx T_{LR}$ as $n \rightarrow \infty$; $J(\theta_0) \approx I(\theta_0)$ as $n \rightarrow \infty$; When $\ell(\theta)$ 是关于 θ 的二次函数 $\Rightarrow$ All same. Geometric: (T_S , 比斜率), (T_W , 比宽度), (T_{LR} , 比高度) ps: $z_{\alpha, n} = \chi_{\alpha, n}^2$				

2.2 For Single Parameter (Numerical Method)

Suppose initial value solution θ_0	Newton-Raphson Method ps: also called Newton's method.	Fisher's Method of Scoring ps: also called scoring method
Formula	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\ell''(\theta_r)} = \theta_r + \frac{U(\theta_r)}{J(\theta_r)}$	$\theta_{r+1} = \theta_r - \frac{U(\theta_r)}{\mathbb{E}[\ell''(\theta_r)]} = \theta_r + \frac{U(\theta_r)}{I(\theta_r)}$
Advantages	Converge <i>very quickly</i> near optimum	I is constant / easier to calculate; Converge for wider θ_0
Stopping Rule	Given tolerance level ε : 1 Stop after r if $ U(\theta_r) \leq \varepsilon$ (Common) or 2 Stop after $ \theta_r - \theta_{r-1} \leq \varepsilon$	

2.3 For Multiple Parameters

*此节中, θ 代表向量, $\theta = (\theta_1, \dots, \theta_p)$ I^{-1} 指其逆矩阵 True Parameter: θ_0

Name	Test Statistic ($T_{Score}^2, T_{Wald}^2, T_{LR}$)	Hypothesis	(Confidence Region) CR
Score Test	$\mathbf{U}(\theta_0)^T I^{-1}(\theta_0) \mathbf{U}(\theta_0) > z_{\alpha, p}$	$\theta = \theta_0$	$\left\{ \theta : \mathbf{U}(\hat{\theta})^T I^{-1}(\theta) \mathbf{U}(\hat{\theta}) < z_{\alpha, p} \right\}$
Wald Test	$(\hat{\theta} - \theta_0)^T I(\theta_0) (\hat{\theta} - \theta_0) > z_{\alpha, p}$	$\theta = \theta_0$	$\left\{ \theta : (\hat{\theta} - \theta)^T I(\theta) (\hat{\theta} - \theta) < z_{\alpha, p} \right\}$
log-Likelihood Ratio Test	$2(\ell(\hat{\theta}) - \ell(\theta_0)) > z_{\alpha, p}$	$\theta = \theta_0$	$\left\{ \theta : 2(\ell(\hat{\theta}) - \ell(\theta)) < z_{\alpha, p} \right\}$
$I(\theta_0) \approx I(\hat{\theta}), J(\theta_0), J(\hat{\theta})$ $ps : z_{\alpha, p} = \chi_{\alpha, p}^2$ T_{LR} : simple hypothesis testing, ($\dim(\mathcal{A}) = 0$) θ 都是向量			

Marginal Normal distribution $\hat{\theta}$:

$\hat{\theta}_j \sim \mathcal{N}(\theta_j, [I^{-1}(\theta_0)]_{jj})$ θ_j : true para

$\cdot ESE(\hat{\theta}_j) = \sqrt{[I^{-1}(\hat{\theta})]_{jj}}$

$\cdot CI(\theta_j): \hat{\theta}_j \pm z_{\alpha} ESE(\hat{\theta}_j)$

Composite Hypothesis Test: $H_0 : \underline{\theta} \in \mathcal{A}, H_1 : \underline{\theta} \notin \mathcal{A}$ reduced model $LR = \frac{\max_{\theta \in \mathcal{A}} L(\theta)}{\max_{\theta \in \Theta} L(\theta)} = \frac{L(\hat{\theta}_{\mathcal{A}})}{L(\hat{\theta}_0)}$ ps: 多维未知参数, H_0 只想其中几个时用 $\| \dim(\mathcal{A})$ 常数时为 0

· **Generalised Likelihood Ratio Test:** $T_{GLR} = -2\ln(LR) = 2(\ell(\hat{\theta}) - \ell(\hat{\theta}_{\mathcal{A}})) \sim \chi_{p-r}^2$ p : # of parameters. $r = \dim(\mathcal{A})$

2.4 For Multiple Parameter (Numerical Method)

Method (start at θ_0)	*此节中, θ 代表向量 Newton-Raphson Method	Fisher's Method of Scoring
Formula	$\theta_{r+1} = \theta_r - \mathbf{H}^{-1}(\theta_r) \mathbf{U}(\theta_r) = \theta_r + \mathbf{J}^{-1}(\theta_r) \mathbf{U}(\theta_r)$	$\theta_{r+1} = \theta_r - (\mathbb{E}[\mathbf{H}(\theta_r)])^{-1} \mathbf{U}(\theta_r) = \theta_r + I^{-1}(\theta_r) \mathbf{U}(\theta_r)$
ps: Jacobian / Hessian $\mathbf{H} = -\mathbf{J}$ $\ $ If $I = \text{diag}(a, b, \dots) \Rightarrow I^{-1} = \text{diag}(a^{-1}, b^{-1}, \dots)$		

3 Bayesian Inference

3.1 Bayesian Approach & Definitions

Basic Idea: Consider θ as a r.v.. Bayesian approach $\Rightarrow$ Probability are used to represent degree of believe about the unknown θ

Prior Distribution $p(\theta)$: Prior distribution $p(\theta)$ contains all available prior information about θ before observing any data.

· **Hyper-Parameters:** 由于 $p(\theta)$ 是已知,服从某种分布 (with parameter γ). We denoted $p(\theta)$ as $p(\theta; \gamma)$ where γ is Hyper-parameters. 定义: $\gamma_{prior} \rightarrow \gamma_{post}$

Likelihood function $p(y|\theta) = f(y|\theta)$: Any observed data is considered as a realisation of r.v. Y with $f(y|\theta)$

Posterior Distribution $p(\theta|y)$: Posterior distribution $p(\theta|y)$, contains all available knowledge.

· **Bayes' Rule / Theorem:** $p(\theta|y) = \frac{p(y|\theta)p(\theta)}{p(y)} = \frac{f(y|\theta)p(\theta)}{m(y)}$ $p(y) = m(y) = \int_{\theta \in \Theta} f(y|\theta)p(\theta)d\theta$ (Normalizing constant)

Name	MAPMaximum A Posteriori	MMSEPosterior Mean / Bayesian Minimum Mean Squared Error Estimator	Posterior Median $\hat{\theta}_{median}$
Single Para	$\hat{\theta}_{MAP} = \arg \max_{\theta} p(\theta y)$	$\hat{\theta}_{MMSE} = \mathbb{E}[\theta y] = \int_{\theta \in \Theta} \theta p(\theta y)d\theta$	$\frac{1}{2} = \int_{-\infty}^{\hat{\theta}_{median}} p(\theta y)d\theta = \int_{\hat{\theta}_{median}}^{\infty} p(\theta y)d\theta$
Name	Posterior Var	$C_{\alpha} = [a, b]$ (1 - α)100% Bayesian Credible Intervals (or Simply Credible Intervals)	
Single Para	$Var[\theta y] = \int_{\theta \in \Theta} (\theta - \mathbb{E}[\theta y])^2 p(\theta y)d\theta$	$\mathbb{P}(a \leq \theta \leq b y) = \int_a^b p(\theta y)d\theta = 1 - \alpha$ ps: $\mathbb{P}(\theta < a y) = \mathbb{P}(\theta > b y) = \alpha/2$	

Bayesian Decision Theory: 目的: ¹Estimating parameter θ ²Test Hypothesis about θ ³Find CI. ps: 后两种有很多种方法来判断

1. Choose a Loss Function $Q(\hat{\theta}, \theta)$ (损失函数, 衡量估计值与真实值的差距; 贝叶斯决策的目的是最小化损失函数的期望)

2. Minimises Posterior Expacted Loss Function: $\hat{\theta}_Q(y) = \arg \min_{\hat{\theta}} \mathbb{E}[Q(\hat{\theta}, \theta)|y] = \arg \min_{\hat{\theta}} \int_{\theta \in \Theta} Q(\hat{\theta}, \theta)p(\theta|y)d\theta$

3.2 Exact Bayesian Inference with Conjugate priors

Conjugate Prior: For $\theta, p(\theta)$ is Conjugate Prior for $p(y|\theta) = f(y|\theta)$ iff 对 $p(\theta)$ 用 Bayes' Rule 得到的 $p(y|\theta)$ 都在同一类型的分布

Conjugate Family: The collection of all Conjugate Prior.

Choice of Prior: Observed data number n : ¹ n small $\Rightarrow$ Posterior shows a great dependency of the choice of prior.

² n middle $\Rightarrow$ posterior less affected by choice of prior. ³ n large $\Rightarrow$ choice of prior has limited effect on posterior.(Likelihood has a strong impact)

Comparing the frequentist and Bayesian approaches: ¹ $\hat{\theta}_{MMSE} \approx \hat{\theta}_{MAP}$ for most cases, especially n large. ² n large: $\hat{\theta}_{MAP} \approx \hat{\theta}_{MLE}$, $CI \approx C_{\alpha}$

3.3 Approximate Bayesian Inference

Intractable Posterior: Non-Conjugate model. (not analytic, no closed-form)

Monte Carlo Methods: Any func $g(\theta) || I_g = \int_{\theta \in \Theta} g(\theta)p(\theta|y)d\theta \approx \frac{1}{M} \sum_{i=1}^M g(\theta^{(i)}) \xrightarrow{a.s.} I_g$ $\theta^{(i)} \sim p(\theta|y), i = 1, ..., M$

1. **Importance Sampling (IS):** ¹ Simulating samples from another distribution $q(\theta)$ (Proposal Distribution) Consistent, Biased for finite

² $w^{(i)} = \frac{\pi(\theta^{(i)}|y)}{q(\theta^{(i)})}$, $\pi(\theta; y) = f(y|\theta)p(\theta)$ ³ IS estimator: $\hat{I}_g^M = \sum_{i=1}^M g(\theta^{(i)})\bar{w}^{(i)} \xrightarrow{a.s.} I_g$, Normalized Weight: $\bar{w}^{(i)} = \frac{w^{(i)}}{\sum_{j=1}^M w^{(j)}}$ $\theta^{(i)} \sim q(\theta)$ ps: $\sum_{j=1}^M \bar{w}^{(j)} = 1$

Remark: Choice of $q(\theta)$ is important. $|| \hat{I}_g^M$ depends on: ¹# of M ; ²mismatch between proposal and the integrand of the integral. Good if $q(\theta) \propto g(\theta)p(\theta|y)$, difficult to achieve.:(

2. **Markov Chain Monte Carlo (MCMC):** MCMC 算法通过 Markov Chain 生成样本, 使得样本接近服从后验分布.(计算量大)

Laplace Approximation: (最简单的估算 posterior 的方法) $p(\theta|y) \approx \mathcal{N}(\underline{\theta}_{MAP}; H^{-1}(\underline{\theta}_{MAP}))$ 类似 MLE 的 J $H(\theta) = -\frac{d^2 \ln(p(\theta|y))}{d\theta^2}$ (适合 $p(\theta|y)$ 接近正态分布的时适用)

Variational Inference (VI): ¹Choose $q(\theta; \eta)$, η 是 q 的参数. ² 选择衡量 $p(\theta|y), q(\theta; \eta)$ 差异的方法 e.g.**KL divergence:** $KL(q(\theta; \eta)||p(\theta|y)) = \int_{\theta \in \Theta} q(\theta; \eta) \ln(\frac{q(\theta; \eta)}{p(\theta|y)})d\theta$

³ 优化参数 (Find parameters to minimise the divergence): $\underline{\eta}^* = \arg \min_{\eta \in A_{\eta}} KL(q(\theta; \eta)||p(\theta|y))$ ⁴ $p(\theta|y) \approx q(\theta; \underline{\eta}^*)$

Remark: Both Laplace Approximation & Variation Inference approximate $p(\theta|y)$ by a parameter family of distributions. $||$ 拉普拉斯是正态分布家族,VI 是更普遍分布家族

4 Linear Model

4.1 Simple Linear Model ($y = \beta x + \varepsilon$)

Model: $Y_i = \mu_i + \varepsilon_i$, $\mu_i = \beta x_i$ $i = 1, 2, ..., n$ $||$ Suppose ¹ ε_i independent ² $\mathbb{E}[\varepsilon_i] = 0$ ³ $Var[\varepsilon_i] = \sigma^2$ $||$ Y_i response variable, x predictor variable.

Least Square (LS) Estimation	RSSResidual Sum of Squares: $S(\beta) = \sum (y_i - \mu_i)^2 = \sum (y_i - x_i \beta)^2$	LS Method: $\hat{\beta}_i = \arg \min_{\beta} S(\beta)$ ps: 求偏导 =0 $ \hat{\beta} = \frac{\sum x_i y_i}{\sum x_i^2}$
$\mathbb{E}, Var$ For β	$\mathbb{E}[\hat{\beta}] = \beta, unbiased Var[\hat{\beta}] = \sigma_{\hat{\beta}}^2 = (\sum x_i^2)^{-1} \sigma^2$ (σ^2 known)	$\widehat{Var}[\hat{\beta}] = \widehat{\sigma}_{\hat{\beta}}^2 = (\sum x_i^2)^{-1} \cdot \widehat{\sigma}^2 \widehat{\sigma}^2 = \frac{1}{n-1} \sum (y - x_i \hat{\beta})^2$
H_0, H_1 Test $ $ CI	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim \mathcal{N}(0, 1)$ (σ^2 known)	$T = (\hat{\beta} - \beta_0) / \widehat{Var}(\hat{\beta}) \sim t_{n-1}$ (σ^2 unknown)
Other	Fitted Values: $\hat{\mu}_i = x_i \hat{\beta} $ Residuals: $\hat{\varepsilon}_i = y_i - \hat{\mu}_i$	For CI / PI / H_0/H_1 Test: Assume $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$
PI	σ^2 know: $\hat{y}_{new} \pm z_{\alpha/2} \sigma \sqrt{1 + \frac{x_0^2}{S_{xx}}}$	σ^2 unknown: $\hat{y}_{new} \pm t_{\alpha/2, n-1} \hat{\sigma}_{\hat{\beta}} \sqrt{1 + \frac{x_0^2}{S_{xx}}}$

4.2 Simple Linear Model ($y = \alpha + \beta x + \varepsilon$)

Model: $Y_i = \mu_i + \varepsilon_i$, $\mu_i = \alpha + \beta x_i$ $i = 1, 2, ..., n$ $||$ Suppose ¹ ε_i independent ² $\mathbb{E}[\varepsilon_i] = 0$ ³ $Var[\varepsilon_i] = \sigma^2$

SL Estimate	$S_{xy} = \frac{1}{n} \sum (x_i - \bar{x})(y_i - \bar{y}) = \sum x_i y_i - \frac{1}{n} (\sum x_i)(\sum y_i)$	LS Method: $\hat{\beta} = \frac{S_{xy}}{S_{xx}} \hat{\alpha} = \bar{y} - \hat{\beta} \bar{x}$	$\hat{\alpha}, \hat{\beta} = \arg \min_{\alpha, \beta} S(\alpha, \beta) S(\alpha, \beta) = \sum (y_i - \mu_i)$
$\mathbb{E}, Var$ For β	$\mathbb{E}[\hat{\beta}] = \beta, unbiased Var[\hat{\beta}] = \sigma_{\hat{\beta}}^2 = \frac{\sigma^2}{S_{xx}}$	$\widehat{Var}[\hat{\beta}] = \widehat{\sigma}_{\hat{\beta}}^2 = \frac{\widehat{\sigma}^2}{S_{xx}}$	$\widehat{\sigma}^2 = \frac{1}{n-2} \sum (y - \alpha - \beta x)^2 = \frac{1}{n-2} (S_{yy} - \frac{S_{xy}^2}{S_{xx}})$
$\mathbb{E}, Var$ For α	$\mathbb{E}[\hat{\alpha}] = \alpha, unbiased Var[\hat{\alpha}] = \sigma_{\hat{\alpha}}^2 = \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$	$\widehat{Var}[\hat{\alpha}] = \widehat{\sigma}_{\hat{\alpha}}^2 = \widehat{\sigma}^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right)$	$\widehat{\sigma}^2 = \frac{1}{n-2} \sum (y - \alpha - \beta x)^2 = \frac{1}{n-2} (S_{yy} - \frac{S_{xy}^2}{S_{xx}})$
H_0, H_1 Test $ $ CI	$T = (\hat{\beta} - \beta_0) / Var(\hat{\beta}) \sim \mathcal{N}(0, 1)$ (σ^2 known)	$T = (\hat{\beta} - \beta_0) / \widehat{Var}(\hat{\beta}) \sim t_{n-2}$ (σ^2 unknown)	$T = (\hat{\alpha} - \alpha_0) / \widehat{Var}(\hat{\alpha}) \sim t_{n-2}$ (σ^2 unknown)
(ctd.) Other	Fitted Values: $\hat{\mu}_i = \hat{\alpha} + \hat{\beta} x_i $ 残差: $\hat{\varepsilon}_i = y_i - \hat{\mu}_i \sum \hat{\varepsilon}_i = 0, \sum \hat{\varepsilon}_i x_i = 0$	For CI / PI / H_0/H_1 Assume $\varepsilon_i \sim \mathcal{N}(0, \sigma^2)$	$T = (\hat{\alpha} - \alpha_0) / Var(\hat{\alpha}) \sim \mathcal{N}(0, 1)$ (σ^2 known)
PI Cor.r = $\frac{S_{xy}}{\sqrt{S_{xx} S_{yy}}}$	σ^2 know: $\hat{y}_{new} \pm z_{\alpha/2} \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$	σ^2 已知: $\hat{y}_{new} \pm t_{\alpha/2, n-2} \sigma \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$	σ^2 未知: $\hat{y}_{new} \pm t_{\alpha/2, n-2} \hat{\sigma}_{\hat{y}} \sqrt{1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}}$

4.3 General Linear Model

Lamma $||E, Var$ in Matrix: If r.v. $\mathbf{v}$, $\mathbf{A}, \mathbf{b}$ constant $\Rightarrow$ ¹ $\mathbb{E}[\mathbf{A}\mathbf{v} + \mathbf{b}] = \mathbf{A}\mathbb{E}[\mathbf{v}] + \mathbf{b}$ ² $Var[\mathbf{A}\mathbf{v} + \mathbf{b}] = \mathbf{A}Var[\mathbf{v}]\mathbf{A}^T$

Metric Predictor: 设计矩阵用连续观测值. $||$ **Factor Predictor:** 离散数据用指示函数 (0/1) 填满, 每个"小组"对应一个指示函数.

Linear Model: $Y_i = \mu_i + \varepsilon_i$, $i = 1, 2, ..., n$ $\mu = f(\underline{\beta}, \mathbf{x}) || f(\underline{\beta}, \mathbf{x})$ linear. $|| \varepsilon_i \stackrel{i.i.d.}{\sim}$ distribution. $\mathbb{E}[\varepsilon_i] = 0$ $Var[\varepsilon_i] = \sigma^2$

· **Matrix Expression:** $\underline{\mathbf{y}} = \mathbf{X}\underline{\beta} + \underline{\varepsilon} ||$ Response Vector: $\underline{\mathbf{y}}$ | Design / Model Matrix: $\mathbf{X}$ | Parameters / Coefficients: $\underline{\beta}$ | Additive Noise Vector: $\underline{\varepsilon}$

· **Parameter Estimates:** $\hat{\underline{\beta}}$ | **Estimated / Fitted Value Mean Vector:** $\hat{\underline{\mu}} = \mathbf{X}\hat{\underline{\beta}}$ | **Estimated / Fitted Residual Vector:** $\hat{\underline{\varepsilon}} = \underline{\mathbf{y}} - \hat{\underline{\mu}}$

Orthonormal Matrix Q	Def: Q := columns and rows are orthonormal vectors.	$Q^T = Q^{-1}$	$QQ^T = Q^TQ = I_n$	$\forall \mathbf{v}, \mathbf{v}' ^2 = Q\mathbf{v}' ^2 = Q^T\mathbf{v}' ^2$
QR Decomposition	$\forall \mathbf{X} \in \mathbb{R}^{n \times p} \Rightarrow \mathbf{X} = Q \begin{pmatrix} \mathbf{R} \\ 0 \end{pmatrix} = \mathbf{QR}$	$\mathbf{R} \in \mathbb{R}^{p \times p}$ is upper triangular matrix; $Q \in \mathbb{R}^{n \times n}$ is orthonormal matrix; Q 前 p 列等于 $\mathbf{Q} \in \mathbb{R}^{n \times p}$		
Least Square (LS)	$RSS := S(\underline{\beta}) = \sum_{i=1}^n \varepsilon_i^2(\underline{\beta}) = \sum_{i=0}^n (y_i - \mu_i(\underline{\beta}))^2 = \mathbf{y} - \underline{\mu}' ^2 = \mathbf{y} - \mathbf{X}\underline{\beta}' ^2 = \underline{\varepsilon}(\underline{\beta}) ^2$			$\hat{\underline{\beta}} = \arg \min_{\underline{\beta}} S(\underline{\beta}) = \arg \min_{\underline{\beta}} \mathbf{f} - \mathbf{R}\underline{\beta}' ^2$
QR Method of LS	Let 1Q be orthonormal matrix ${}^2\mathbf{X} = \mathbf{QR}$ ${}^3Q^T\mathbf{y} = \begin{pmatrix} \mathbf{f} \in \mathbb{R}^p \\ \mathbf{r} \in \mathbb{R}^{n-p} \end{pmatrix} \Rightarrow S(\underline{\beta}) = Q^T \underline{\varepsilon}(\underline{\beta}) ^2 = \dots = \mathbf{f} - \mathbf{R}\underline{\beta}' ^2 + \mathbf{r}' ^2$			
QR Method (ctd.)	If $\mathbf{R}$ 满秩: $\hat{\underline{\beta}} = \mathbf{R}^{-1}\mathbf{f} = \mathbf{R}^{-1}Q^T\mathbf{y}$	$RSS = \mathbf{r}' ^2 = \underline{\varepsilon}(\hat{\underline{\beta}}) ^2 = \mathbf{y} - \mathbf{X}\hat{\underline{\beta}}' ^2$	$RSE = \frac{RSS}{n-p} = \hat{\sigma}^2$	$\mathbf{ps}:\mathbf{f} = Q^T\mathbf{y} \quad \quad Q^TQ = I_p, QQ^T \neq I$

Matrix Method of LS	$S(\underline{\beta}) = (\mathbf{y} - \mathbf{X}\underline{\beta})^T(\mathbf{y} - \mathbf{X}\underline{\beta}) = \mathbf{y}^T\mathbf{y} - 2\mathbf{y}^T\mathbf{X}\underline{\beta} + \underline{\beta}^T\mathbf{X}^T\mathbf{X}\underline{\beta}$	$\underline{\hat{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}$	$\mathbf{V}_{\underline{\hat{\beta}}} = Var[\underline{\hat{\beta}}] = (\mathbf{X}^T\mathbf{X})^{-1}\sigma^2$
Influence / Hat Matrix A	$\mathbf{A} = \mathbf{Q}\mathbf{Q}^T = \mathbf{A}^2 = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$	# of (identifiable) parameters = $tr(\mathbf{A}) = tr(\mathbf{Q}\mathbf{Q}^T) = tr(I_p) = p$	
$Var[\mathbf{y}], Var[\mathbf{f}], \underline{\hat{\mu}}, \underline{\hat{\beta}}$	$Var[\mathbf{y}] = Cov(\underline{\varepsilon}, \underline{\varepsilon}) = \sigma^2 I_n$	$Var[\mathbf{f}] = \mathbf{Q}^T\mathbf{Q}\sigma^2 = I_p$	$\underline{\hat{\mu}} = \mathbf{X}\underline{\hat{\beta}} = \mathbf{A}\mathbf{y}$
$\mathbf{V}_{\underline{\hat{\beta}}}, \mathbf{V}_{\underline{\hat{\varepsilon}}}, \mathbf{V}_{\underline{\hat{\mu}}} \parallel \underline{\hat{\mu}} \text{ dist}$	$\mathbf{V}_{\underline{\hat{\beta}}} = Var[\underline{\hat{\beta}}] = \mathbf{R}^{-1}\mathbf{R}^{-T}\sigma^2$	$\mathbf{V}_{\underline{\hat{\varepsilon}}} = Var[\underline{\hat{\varepsilon}}] = (I_n - \mathbf{A})\sigma^2$	$\mathbf{V}_{\underline{\hat{\mu}}} = Var[\underline{\hat{\mu}}] = \mathbf{A}\sigma^2$
$\mathbb{E}[\underline{\hat{\beta}}], \mathbb{E}[\underline{\hat{\varepsilon}}], \mathbb{E}[\underline{\hat{\mu}}] \parallel \underline{\hat{\varepsilon}} \text{ dist}$	$\mathbb{E}[\underline{\hat{\beta}}] = \underline{\beta} \text{ unbiased}$	$\mathbb{E}[\underline{\hat{\varepsilon}}] = \mathbf{0}$	$\mathbb{E}[\underline{\hat{\mu}}] = \underline{\mu} \text{ (unbiased)}$
If $\mathbf{X}^T\mathbf{X}$ is diagonal $\Rightarrow \mathbf{X}$ is called Orthogonal. $\Rightarrow$ In this Case: $\mathbf{V}_{\underline{\hat{\beta}}}$ diagonal $\Rightarrow \beta_i$ independent.			

$\mathbf{V}_{Q^T \mathbf{y}}, \mathbb{E}[Q^T \mathbf{y}] \parallel \hat{\sigma}^2, \hat{\mathbf{V}}_{\hat{\beta}}$	$\mathbf{V}_{Q^T \mathbf{y}} = Var[Q^T \mathbf{y}] = I_n \sigma^2$	$\mathbb{E}[Q^T \mathbf{y}] = \mathbb{E}\left[\begin{pmatrix} \mathbf{f} \\ \mathbf{r} \end{pmatrix}\right] = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix} \underline{\beta}$	$\hat{\sigma}^2 = \frac{\ \mathbf{r}\ ^2}{n-p}$ (unbiased)	$\hat{\mathbf{V}}_{\hat{\beta}} = \mathbf{R}^{-1} \mathbf{R}^{-T} \hat{\sigma}^2$ (unbiased)	$\parallel \hat{\beta}, \hat{\sigma}^2$ independent
$\mathbf{f}, \mathbf{r}$ Distribution	Known σ^2 : $\mathbf{f} \sim \mathcal{N}(\mathbf{R}\underline{\beta}, I_p \sigma^2)$ $\mathbf{r} \sim \mathcal{N}(\mathbf{0}, I_{n-p} \sigma^2)$		Unknown σ^2 : $\hat{\sigma}_{\beta_i} = \sigma_{\beta_i} \hat{\sigma} / \sigma \parallel \hat{\sigma}_{\beta_i} = \sqrt{[\hat{\mathbf{V}}_{\hat{\beta}}]_{ii}}$ $\sigma_{\beta_i} = \sqrt{[\mathbf{V}_{\underline{\beta}}]_{ii}}$		
Test Statistic	For $\underline{\beta}$ 整体: $T_{known \sigma^2}(\mathbf{Y}) = \frac{1}{\hat{\sigma}^2} \ \mathbf{r}\ ^2 = \frac{1}{\hat{\sigma}^2} \sum_{i=1}^{n-p} r_i^2 \sim \chi_{n-p}^2$		For β_i 个体: $T_{unknown \sigma^2}(\mathbf{Y}) = (\hat{\beta}_i - \beta_i) / \hat{\sigma}_{\beta_i} \sim t_{n-p}$		

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Beta $X \sim \text{Beta}(\alpha, \beta), \alpha, \beta > 0$	$\frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha, \beta)}, x \in [0, 1]$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	$B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}$
When $\text{Beta}(\alpha, \beta)$ has maximum probability, $x = (\alpha - 1)/(\alpha + \beta - 2)$				
Gamma $X \sim \Gamma(\alpha, \beta), \alpha, \beta > 0$	$\frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, x \geq 0$	$\frac{\alpha}{\beta}$	$\frac{\alpha}{\beta^2}$	$\Gamma(\alpha) = (\alpha - 1)!$ $\Gamma(\alpha) = \int_0^\infty x^{\alpha-1} e^{-x} dx$
Student's t $X \sim t_v, v > 0$	$\frac{1}{\sqrt{v}B(\frac{v}{2}, \frac{1}{2})} (1 + \frac{x^2}{v})^{-\frac{v+1}{2}}$	0	$\frac{v}{v-2}$	If $Z \sim N(0, 1), Y \sim \chi_n^2$ $\frac{Z}{\sqrt{Y/n}} \sim t_n$
PDF Symmetric at zero and has more mass in the tails (compare to the standard normal distribution)				
When $n \rightarrow \infty$ it will converges to $N(0, 1)$. Cauchy Distribution: $X \sim t_1$ and its mean and var are undefined.				
Chi-Squared $X \sim \chi_v^2, v > 0$	$\frac{1}{2^{\frac{v}{2}} \Gamma(\frac{v}{2})} x^{\frac{v}{2}-1} e^{-\frac{x}{2}}, x \geq 0$	v	$2v$	-
If $X_i \sim N(0, 1)$, then $\sum X_i^2 \sim \chi_n^2$ If $X \sim \chi_n^2, Y \sim \chi_m^2$, then $X + Y \sim \chi_{n+m}^2$				
F $X \sim F(\lambda, v), v > 0$ ps: $f_{a,b;\alpha} = 1/f_{a,b;1-\alpha}$	$\frac{\lambda^{\frac{\lambda}{2}} v^{\frac{v}{2}}}{B(\frac{\lambda}{2}, \frac{v}{2})} x^{\frac{\lambda}{2}-1} (\lambda x + v)^{-\frac{\lambda+v}{2}}, x \geq 0$	$\frac{v}{v-2}$	$\frac{2v^2(\lambda+v-2)}{\lambda(v-2)^2(v-4)}, v > 4$	$Y_1, Y_2 \sim \chi_{n_1}^2, \chi_{n_2}^2, X = \frac{Y_1/n_1}{Y_2/n_2}$ $X \sim F_{n_1, n_2}, X^{-1} \sim F_{n_2, n_1}$

6 Review

6.1 Basic Concepts

Law of Total Probability: $\mathbb{P}(A) = \sum_n \mathbb{P}(A|B_n)\mathbb{P}(B_n)$, $\cup_n B_n = \Omega$, B_n disjoint

Type	Type I error	Type II error	-	H_0 is T	H_0 is F
Def	$\mathbb{P}(\text{Rej } H_0 H_0 \text{ is } T)$	$\mathbb{P}(\text{Fail Rej } H_0 H_0 \text{ is } F)$	Fail to Rej H_0	Right	Type II (β) 与 size 有关
Significance Level: $\alpha = \mathbb{P}(\text{Rej } H_0 H_0 \text{ is } T)$			Rej H_0	Type I (α) 与 size 无关	Right (Power)

P-value: 1.One-side: $p = \mathbb{P}(T \geq t; T \sim H_0)$ 2.Two-sides: $p = 2 \min\{\mathbb{P}(T \geq t; T \sim H_0), \mathbb{P}(T \leq t; T \sim H_0)\}$

Power of Test: $\mathbb{P}(\text{Rej } H_0|H_0 \text{ is } F) = 1 - \beta = \mathbb{P}(Z \in C|\mu = \mu^*)$ · 1.Sample size ↑, Power ↑; 2.σ² ↓, Power ↑; 3.Effect size $|\theta_0 - \theta^*|$ ↑, Power ↑; 4.α↑, Power ↑.

Def of Confidence Interval (CI): 100(1 - α)% Confidence Interval for θ is $[l(X), u(X)]$, $\mathbb{P}(l(X) < \theta < u(X)) = 1 - \alpha$.

(Convergence of r.v.) Name	Definition	written as	Example	
Convergence in Probability	If $\lim F_{X_n}(x) = F_X(x)$ for all $x \in \mathbb{R}$.	$X_n \xrightarrow{d} X$ or $X_n \xrightarrow{d} F_X$	CLT 最弱:	见 下面
Convergence in Distribution	If $\forall \epsilon > 0, \lim \mathbb{P}(X_n - X > \epsilon) = 0$.	$X_n \xrightarrow{p} X$	WLLN 中等:	If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$. $\Rightarrow \bar{X}_n \xrightarrow{p} \mu$
Almost Sure Convergence	If $\mathbb{P}(\lim X_n = X) = 1$	$X_n \xrightarrow{a.s.} X$	SLLN 最强:	If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$, $Var[X_i] < \infty. \Rightarrow \bar{X}_n \xrightarrow{a.s.} \mu$

Central Limit Theorem (CLT): If $X_i \stackrel{i.i.d.}{\sim} F_X, \mathbb{E}[X_i] = \mu$ and $Var[X_i] < \infty. \Rightarrow \frac{\bar{X}_n - \mu}{\sqrt{\sigma^2/n}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} N(0, 1)$ 样本均值的分布会趋近于正态分布

6.2 Test Statistics and Decision Rule

One Sample		Two Sample		
Find μ		Find μ		Find σ^2
Z-Test (σ^2 known) $Z = \frac{\bar{X} - \mu_0}{\sqrt{\sigma^2/n}} \sim N(0, 1)$	t-Test (σ^2 unknown) $T = \frac{\bar{X} - \mu_0}{\sqrt{S^2/n}} \sim t_{n-1}$	Paired t-Test (X,Y same Unit) $D_i = Y_i - X_i \sim N(\mu_D, \sigma_D^2)$ $T = \frac{\bar{D} - \mu_D}{\sqrt{S^2/n}} \sim t_{n-1}$	Two-Sample t-Test (独立) $T = \frac{(\bar{X} - \bar{Y}) - \Delta_0}{\sqrt{\sigma^2(\frac{1}{m} + \frac{1}{n})}} \sim N(0, 1)$ 未知: $S_p^2; T \sim t_{m+n-2}$	F-Test $F = \frac{S_X^2}{S_Y^2} \sim F_{m-1, n-1}$

· 对于 Two-Sample t-Test 我们要求 X,Y σ² 一致 ps: $H_0 : \mu_D = \mu_X - \mu_Y ; H_0 : \Delta_0 = \mu_X - \mu_Y ; X : m, Y : n$

Pooled Sample Variance: (Two sample Variance) $S_p^2 = \frac{(m-1)S_X^2 + (n-1)S_Y^2}{m+n-2}$ ps:Unbiased

7 Writting Format

Distribution: 产生新分布的一种方法: 通过条件概率来. $\mathbb{P}(A|B) = \mathbb{P}(A \cap B)/B$

p-value: p-value measures the plausibility that the data were generated under H_0 . **Low p-value:** low p-value suggests H_0 is incompatible with the data and hence likely to be false.

CI: The set of values (e.g. θ) are not rejected at significance level α under H_0 . If data are generated under H_0 , there is a probability $(1 - \alpha)$ for the true θ_0 to be within the $(1 - \alpha)100\%$ CI.

Hypothesis Testing: Steps: ¹ State H_0, H_1 ; ² Select Test Statistic T ; ³ distribution of T under H_0 ; ⁴ observed test statistic; ⁵ critical value / region, p-value, or CI; ⁶ conclusion. || **Reject H_0 :** We would reject H_0 in favour of the H_1 . (So that ...)

Fail to reject H_0 : We fail to reject H_0 . Hence, there is strong evidence against the null hypothesis.

MLE: Using (Compare) Newton's Method = (≠) Scoring Method: Because the second derivative of likelihood function doesn't (does) include the r.v.

MLE Inference Parameter $|\mathbb{P}(\theta < \gamma) = \alpha$ is false: We cannot make such probability statement. $\{\theta\}$ is a fixed value (but unknown).

Geometric Comparison of $T_{Score}, T_{Wald}, T_{LR}$: **1.TScore** (Slope Comparison): Represents the slope of the likelihood function at θ_0 ; A steeper slope indicates that the likelihood function is changing more rapidly at θ_0 , which implies a greater difference between $\hat{\theta}$ and θ_0 . || **2.TWald** (Width Comparison): Compares the difference between $\hat{\theta}$ and θ_0 . || **3.TLR** (Height Comparison): Compares the maximum likelihood value $L(\hat{\theta})$ and the likelihood value at $\theta_0, L(\theta_0)$. A larger height difference indicates a greater discrepancy between $\hat{\theta}$ and θ_0 || **$T_{Score}, T_{Wald}, T_{LR}$ Other** 部分公式可以近似的理由: This is due to $\hat{\theta}$ being a consistent estimator of θ_0 , and $J(\theta_0)$ being a consistent estimator of $I(\theta_0)$

Normal Linear Regression Assumptions: Assumptions are that the observations: 1. are from a normal distribution; 2. are independent; 3. have a common variance; 4. Linearity of the expectation variable.

Linear Model: LS Estimation: $S(\beta)$ small: μ_i relatively close to the y_i ; Large: μ_i far from their corresponding y_i . || **Meaning of ϵ :** The random component of the model is intended to capture the fact that if we gathered a replicate set of data, the β_i (coefficient) would not change, but the Y_i would, as a result of different measurement errors. || **Why Use Least Square to estimate β :** Most of time, Linear Model can be a system of n equations with p parameters with $n > p$. In this case, the system is overdetermined and there is no exact solution. The least square method is used to find the best solution. || **Geometric Interpretation of Linear Model|LS Method:** The least Square Estimation of Linear Model amounts to finding the *orthogonal projection* of the n dimensional response data y onto the p dimensional linear subspace spanned by the columns of X . || **Geometric Interpretation of Linear Model|E[y]:** E[y] lies in the space spanned by the columns of X , and the least square estimate seeks to find the point in this space that is closest to y . || **Independent of $\hat{\beta}, \hat{\sigma}^2$:** f, r independent $\Rightarrow \hat{\beta}, \hat{\sigma}^2$ independent.

Meaning of R^2 : R^2 evaluates how well the model fits the response data. || 后面减去的部分: proportion of the variance in the response variable that is not explained by the model. || $\rightarrow 1$: A value close to 1 indicates that the model fits well and can explain most of the variability. || $\rightarrow 0$: a low R^2 value does *not necessarily* indicate a poor model; it may simply mean that the data contains a substantial random component.

For Model of Factors: If it's a model of Factors, it's often difficult to interpret the p-value for individual coefficients in any meaningful way. $\Rightarrow$ Prefer to test the whole factor effect.